A type-II_∞ modular-shadow algebra for the Bianchi IX vacuum sector via Manin–Marcolli symbolic dynamics

(conjecture paper, with falsifier protocol)

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Abstract

We propose a type-II_∞ modular-shadow algebra $\mathcal{M}_{\mathrm{BIX}}^{\mathrm{II}_\infty}$ for the vacuum sector of Bianchi IX cosmology in the Belinski–Khalatnikov–Lifshitz (BKL) mixmaster phase, obtained as the Speranza (2025) crossed product of a wedge-complement type-III_1 field algebra $\mathcal{N}$ by the Connes–Takesaki modular automorphism flow. Combining Manin–Marcolli's (2015) symbolic-dynamics representation of the BKL Kasner bounce sequence as the geodesic flow of the modular surface $X(2) = \Gamma(2)\backslash\mathbb{H}$ with the generic crossed-product framework of Witten (2022) and Speranza (2025), we *conjecture* that the modular flow rate on $\mathcal{M}_{\mathrm{BIX}}^{\mathrm{II}_\infty}$ equals $\lambda_{\mathrm{mod}} = 2\pi\,\lambda_{\mathrm{BKL}}$, where $\lambda_{\mathrm{BKL}} = h_{\mathrm{KS}}(\sigma_{\mathrm{Gauss}})$ is the Kolmogorov–Sinai entropy of the Gauss-shift symbolic encoding of BKL Kasner bounces. This in turn predicts a Maldacena–Shenker–Stanford-style modular Krylov-complexity bound $\mathcal{C}_K^{\mathrm{mod}}(s) \le A\,\cosh(2\pi\,\lambda_{\mathrm{BKL}}\,s)$ on $\mathcal{M}_{\mathrm{BIX}}^{\mathrm{II}_\infty}$. The construction is presented as four explicit conjectures (Conjectures M45.1.A–M45.1.D) and a one-sided numerical falsifier based on direct computation of λ_{BKL} via the Gauss shift and comparison with Krylov-rate growth in a discretised Bianchi IX wave-function evolution. We do *not* prove any of the four conjectures; six explicit "[TBD: prove]" markers are flagged. The proposal is compatible with the Heinzle–Uggla (2009) Bianchi IX attractor theorem and with our companion work on the Modular Shadow MSS bound for the isotropic FLRW sector.

1 Introduction

The Maldacena–Shenker–Stanford (MSS) bound on chaos [1] states that the Lyapunov rate of any large-N thermal quantum system satisfies $\lambda_L \le 2\pi/\beta$. Its modular-Krylov refinement of Caputa–Magán–Patramanis–Tonni [3] and the universal operator-growth inequality of Parker–Cao–Avdoshkin–Scaffidi–Altman [2] give a sharper, large-N-free statement on the operator-Hilbert space of modular evolution.

In parallel, the program initiated by Witten [4, 5] and Chandrasekaran–Longo–Penington–Witten [6] has established that gravitational observable algebras with quantum reference frames are type-II_∞ crossed products $\mathcal{A}_{\mathrm{II}_\infty} = \mathcal{A}_{\mathrm{III}_1} \rtimes_{\sigma^\phi} \mathbb{R}$, restoring a semifinite trace.

Speranza [8] extended this to a generic *intrinsic* cosmological observer construction, applicable to backgrounds without a global timelike Killing vector. Kudler-Flam–Leutheusser–Satishchandran [7] concurrently worked out the FLRW (isotropic) case in detail.

The Bianchi IX vacuum, with its anisotropic SO(3)-symmetric metric and three independent scale factors, is the canonical *anisotropic* analog of FLRW. In the BKL approach to its initial-singularity behaviour, generic solutions exhibit a chaotic infinite sequence of Kasner epochs separated by bounces governed by a hyperbolic billiard in Misner variables [11, 12]. Manin–Marcolli [9] proved a remarkable correspondence:

$$\text{Mixmaster BKL bounce sequence} \quad \longleftrightarrow \quad \text{geodesic flow on } X(2) = \Gamma(2)\backslash\mathbb{H}, \tag{1}$$

realised by a two-sided continued-fraction shift, with Gauss measure as the unique invariant probability of maximal entropy. Fan–Fathizadeh–Marcolli [10] further showed that the Connes spectral action on Bianchi IX gravitational instantons admits modular-form expansions of weights 2, 14, 18.

Aim of this paper. We propose a *conjectural* synthesis of the two pictures: combine the Speranza/CLPW crossed-product construction with the Manin–Marcolli geodesic-flow representation of BKL chaos, identify the resulting type-II_∞ algebra $\mathcal{M}_{\mathrm{BIX}}^{\mathrm{II}_\infty}$ as the *modular-shadow algebra* of the Bianchi IX vacuum sector, and *conjecture* that its modular flow rate equals 2π times the Kolmogorov–Sinai entropy λ_{BKL} of the Gauss shift. This in turn predicts an MSS-style Krylov-complexity bound at $\beta_{\mathrm{eff}} = 2\pi/\lambda_{\mathrm{BKL}}$.

What this paper does *not* prove. None of the four sub-conjectures (Conjectures 2–6) is proved. Six explicit "[TBD: prove]" markers are flagged in the construction sketch (§4). This is a *programmatic* note: it identifies a precise, testable target, sketches the construction, and presents a one-sided numerical falsifier (§7).

Honesty bar. The Manin–Marcolli correspondence (1) is a theorem; the type-II_∞ crossed-product framework of Witten/Speranza/CLPW is established; the MSS bound and its Krylov refinement are established. The *novelty* here lies entirely in identifying the rate of the Speranza modular flow with $2\pi\lambda_{\mathrm{BKL}}$. That identification is open.

Organization. §2 reviews the three preliminary frameworks. §3 states the central Conjecture M45.1. §4 sketches the four-step construction. §5 formulates the MSS-style Krylov bound. §6 relates to our Bianchi IX Lemma A.1 (paper #5). §7 presents a numerical falsifier protocol. §8 lists open problems.

2 Preliminaries: three frameworks

2.1 Manin–Marcolli BKL ↔ X(2) geodesic flow

Misner variables for Bianchi IX vacuum reduce the dynamics to the geodesic flow of an indefinite metric on a hyperbolic billiard table in $\mathbb{H}^2$ [11]. Cornish–Levin [12] demonstrated, via observer-independent fractal methods, the existence of a strange repellor in the mixmaster dynamics. Manin–Marcolli [9] showed that the BKL Kasner-bounce sequence is conjugate to the two-sided shift $\sigma : \Sigma_2 \to \Sigma_2$ associated with the continued-fraction expansion, equivalently the geodesic flow on the modular surface $X(2) = \Gamma(2)\backslash\mathbb{H}$ along its

imaginary axis. The unique invariant probability of maximal entropy for σ is the Gauss measure

$$d\mu_{\text{Gauss}}(x) = \frac{1}{\log 2}\,\frac{dx}{1+x}, \qquad x \in [0,1], \tag{2}$$

with topological entropy $h_{\text{top}}(\sigma) = \pi^2/6$ (the "Lochs–Khinchin" constant; see [9] §2.4).

Definition 1 (BKL Lyapunov rate)**.** *We denote by*

$$\lambda_{\text{BKL}} \;:=\; h_{\text{KS}}(\sigma, \mu_{\text{Gauss}}) \;=\; \frac{\pi^2}{6\log 2} \;\approx\; 2.3731, \tag{3}$$

the Kolmogorov–Sinai entropy of the Gauss shift with respect to μ_{Gauss} (Manin–Marcolli [9], Theorem 2.4 and [10]).

2.2 Witten II_∞ crossed product

Let $\mathcal{N}$ be the type-III_1 von Neumann algebra of gauge-invariant local field operators on a wedge-complement region of some background spacetime, in the cyclic-and-separating GNS representation of a KMS state ϕ at inverse temperature β. Tomita–Takesaki gives the modular automorphism group $\sigma_t^\phi : \mathcal{N} \to \mathcal{N}$. Witten [5] and CLPW [6] showed that the crossed product $\mathcal{N} \rtimes_{\sigma^\phi} \mathbb{R}$ is type-II_∞, semifinite, with trace dual to the modular flow. The Connes–Takesaki flow of weights identifies the natural rate of σ_t^ϕ with $2\pi/\beta$ (in normalised units).

2.3 Speranza intrinsic observer

For backgrounds without a global timelike Killing vector (de Sitter static patch, FLRW, Bianchi), Speranza [8] constructs an *intrinsic cosmological observer*: a worldline γ equipped with a quantum reference clock, and an associated crossed-product algebra

$$\mathcal{A}_\gamma^{\text{II}_\infty} \;=\; \mathcal{A}_{\gamma^c}^{\text{III}_1} \rtimes_{\sigma^{\phi_\gamma}} \mathbb{R}, \tag{4}$$

where $\mathcal{A}_{\gamma^c}^{\text{III}_1}$ is the type-III_1 algebra of fields on the wedge complement of γ, and ϕ_γ is the geometric (vacuum-like) state for which a local modular automorphism is well-defined. Kudler-Flam–Leutheusser–Satishchandran [7] worked this out explicitly for FLRW backgrounds. For Bianchi IX, the construction has not yet been carried out; this gap is precisely what Conjectures M45.1.B and M45.1.C below address.

3 Central conjecture

Let M_{BIX} denote a Bianchi IX vacuum spacetime in mixmaster (BKL) phase, γ an intrinsic Speranza observer with worldline parametrised by Misner volume time τ, and $\mathcal{A}_{\text{obs}} := \mathcal{A}_{\gamma^c}^{\text{III}_1}$ the type-III_1 algebra of bounded gauge-invariant local field observables on the wedge complement of γ in M_{BIX}.

Conjecture 1 (Bianchi IX modular-shadow algebra, Conjecture M45.1)**.** *There exists a faithful normal representation $\pi : \mathcal{A}_{\text{obs}} \to B(\mathcal{H}_{\text{BIX}})$ on a Hilbert space $\mathcal{H}_{\text{BIX}}$ unitarily equivalent to $L^2(X(2), \mu_{\text{Gauss}}) \otimes \ell^2$, such that:*

(i) the Speranza crossed product $\mathcal{M}_{\text{BIX}}^{\text{II}_\infty} := \mathcal{A}_{\text{obs}} \rtimes_{\sigma^{\phi_\gamma}} \mathbb{R}$ is a type-II_∞ factor with semifinite trace τ_{II};

(ii) the modular automorphism $\sigma_t^{\phi_\gamma}$ is implemented (up to inner automorphism) by the Manin–Marcolli geodesic flow g_t on $L^2(X(2), \mu_{\text{Gauss}})$;

(iii) consequently, the natural modular flow rate is $\lambda_{\text{mod}} = 2\pi\, \lambda_{\text{BKL}}$ with λ_{BKL} from Definition 1.

We split this into four sub-conjectures (§4) and derive the falsifiable Krylov bound (Conjecture 6, §5).

4 Sketch of construction

4.1 Step A: Wedge algebra setup

Choose an intrinsic observer γ in M_{BIX} with worldline parametrised by Misner volume time τ. The wedge complement γ^c is the spacelike-separated region relative to γ (well-defined locally, since SO(3) symmetry provides a foliation by homogeneous slices [13]).

Conjecture 2 (Wedge algebra type, M45.1.A)**.** *The algebra $\mathcal{A}_{\text{obs}}$ of bounded gauge-invariant local field observables on γ^c is type-III_1, in analogy with the Bisognano–Wichmann theorem for Rindler wedges and the Kudler-Flam–Leutheusser–Satishchandran result for FLRW [7].*

[**TBD-1: prove**] The Bisognano–Wichmann argument uses the global boost Killing vector of Minkowski space, which is absent in Bianchi IX. The KLS argument uses the FLRW conformal-time global slicing, also absent. A new argument is needed, plausibly via Buchholz-Junglas split property combined with the SO(3) homogeneity of Bianchi IX slices.

4.2 Step B: Crossed product

Given Conjecture 2, Tomita–Takesaki provides the modular automorphism $\sigma_t^{\phi_\gamma}$ for any KMS state ϕ_γ, e.g., the Hartle–Hawking-analogue state on M_{BIX} (whose existence is itself non-trivial in Bianchi IX; see [13]).

Conjecture 3 (Speranza transfer, M45.1.B)**.** *The Speranza intrinsic-observer crossed-product construction* (4) *applies to M_{BIX} with τ-foliation, yielding a type-II_∞ factor $\mathcal{M}_{\text{BIX}}^{\text{II}_\infty} = \mathcal{A}_{\text{obs}} \rtimes_{\sigma^{\phi_\gamma}} \mathbb{R}$ with semifinite trace τ_{II}.*

[**TBD-2: prove**] Speranza's flow-of-weights argument uses cosmological time. Bianchi IX has no global cosmological time: only the Misner volume $\tau = -\frac{1}{3}\log V$ plays a comparable role, and τ is monotonic only in the BKL backward direction. The argument must handle the non-globality of τ.

4.3 Step C: Modular-flow = Manin–Marcolli geodesic flow

This is the load-bearing identification.

Conjecture 4 (Flow identification, M45.1.C)**.** *On $\mathcal{H}_{\text{BIX}} \cong L^2(X(2), \mu_{\text{Gauss}}) \otimes \ell^2$, the modular flow $\sigma_t^{\phi_\gamma}$ of Conjecture 3 acts (up to inner automorphism on the ℓ^2 factor) as*

$$\sigma_t^{\phi_\gamma}(f) \;=\; f \circ g_{-ct}, \qquad f \in L^2(X(2), \mu_{\text{Gauss}}), \tag{5}$$

where g_t is the Manin–Marcolli geodesic flow on $X(2)$ and $c > 0$ is a normalisation constant identified below.

[**TBD-3: prove**] The identification of the Speranza-Connes-Takesaki flow with the Manin–Marcolli geodesic flow is the deepest open step. A possible route: realise $\mathcal{A}_{\text{obs}}$ as a sub-algebra of the Bost–Connes type arithmetic algebra associated with $\Gamma(2)$ (Connes–Marcolli *Noncommutative Geometry, Quantum Fields and Motives*, Ch. III), where the modular flow has an intrinsic geometric realisation by the geodesic flow on $X(2)$.

4.4 Step D: Modular-flow rate

Given Conjecture 4, the rate of $\sigma_t^{\phi_\gamma}$ relative to the natural Gauss-measure parametrisation is $\lambda_{\text{mod}} = 2\pi/\beta_{\text{eff}}$ where β_{eff} is the inverse Hawking-like temperature of the Bianchi IX intrinsic observer.

Conjecture 5 (Rate identification, M45.1.D)**.** *The normalisation constant c in* (5) *is fixed by*

$$\lambda_{\text{mod}} \;=\; 2\pi\,\lambda_{\text{BKL}} \;=\; \frac{\pi^3}{3\log 2} \;\approx\; 14.907 \quad \textit{(natural BKL units)}, \tag{6}$$

where $\lambda_{\text{BKL}} = h_{\text{KS}}(\sigma_{\text{Gauss}}) = \pi^2/(6\log 2)$ from Definition 1.

[**TBD-4: prove**] The factor 2π comes from the Bisognano–Wichmann normalisation $\beta_{\text{BW}} = 2\pi$ in the absence of explicit boost generator.

5 Modular Krylov complexity bound

Combining Conjectures M45.1.A–M45.1.D with the modular-Krylov formalism of Caputa–Magán–Patramanis–Tonni [3] and the universal Lanczos-cosh bound of Parker–Cao–Avdoshkin–Scaffidi–Altman [2] yields a falsifiable target.

Conjecture 6 (Bianchi IX modular Krylov bound)**.** *For any operator $\mathcal{O} \in \mathcal{M}_{\text{BIX}}^{\text{II}_\infty}$ with finite modular two-point function $(\mathcal{O},\mathcal{O})_{\phi_\gamma} < \infty$, the modular Krylov complexity satisfies*

$$\mathcal{C}_K^{\text{mod}}(\mathcal{O}; s) \;\le\; A_{\mathcal{O}}\,\cosh\!\big(2\pi\,\lambda_{\text{BKL}}\,s\big), \tag{7}$$

with $A_{\mathcal{O}}$ a state-dependent prefactor and $\lambda_{\text{BKL}} = \pi^2/(6\log 2)$. The corresponding modular Lyapunov exponent is bounded by $\lambda_K^{\text{mod}} \le 2\pi\,\lambda_{\text{BKL}}$.

[**TBD-5: prove**] The Parker–Cao argument in [2] requires linearly growing Lanczos coefficients $b_n \sim \alpha n$. Their argument transposes from Hamiltonian to modular evolution (this is the Caputa et al. [3] step) but the *slope* $\alpha = \pi\lambda_{\text{BKL}}$ is conjectural.

Counter-example caveat. Bhattacharjee–Cao–Nandy–Pathak [14] exhibit saddle-dominated scrambling systems where Krylov complexity grows exponentially without underlying chaos. In Bianchi IX, the BKL stochasticity is genuine (Gauss-shift entropy is positive); but the *converse* statement — saturation of (7) implies BKL chaos — is *not* claimed here.

Effective temperature. Define the effective inverse temperature of the Bianchi IX intrinsic observer by

$$\beta_{\text{eff}} := \frac{2\pi}{\lambda_{\text{BKL}}} = \frac{12\log 2}{\pi} \approx 2.6479. \tag{8}$$

This is a *prediction*: the intrinsic Hawking-like temperature of an observer in the BKL phase of Bianchi IX is $T_{\text{eff}} = 1/\beta_{\text{eff}} \approx 0.3777$ in natural BKL units. The von Neumann coupling constant of $\mathcal{M}_{\text{BIX}}^{\text{II}_\infty}$ (in the Murray–von Neumann sense) is conjecturally proportional to $\lambda_{\text{BKL}}^{-1}$.
[**TBD-6: prove**] The proportionality constant in the von Neumann coupling identification is undetermined and constitutes the main content of the rate conjecture beyond the symbolic-dynamics input.

6 Connection to Lemma A.1 (paper #5)

In [15] we proved (Lemma A.1) that the set of configurations in Bianchi IX vacuum with eigenvalue crossings of the spatial Ricci tensor has Lebesgue measure zero in the Misner configuration space. This is consistent with Conjecture M45.1: eigenvalue crossings correspond to non-generic Kasner-bounce orbits, which form a measure-zero subset of $X(2)$ under μ_{Gauss} (the periodic orbits of σ, dense but countable).

Compatibility check. The Fan–Fathizadeh–Marcolli [10] expansion of the Connes spectral action on Bianchi IX in modular forms of weight $2, 14, 18$ should restrict on $\mathcal{M}_{\text{BIX}}^{\text{II}_\infty}$ to a graded trace-class component; this is a non-trivial consistency check on Conjecture M45.1.

7 Falsifier protocol

The construction above is falsifiable. A concrete numerical protocol (developed in detail in the companion note `falsifier_protocol.md`) consists of:

(F1) *Numerical verification of* λ_{BKL}. Iterate the Gauss shift $\sigma : [0,1] \to [0,1]$, $\sigma(x) = \{1/x\}$, on $N = 10^6$ trajectories of length $T = 10^4$, sample the Lyapunov rate via finite-time Birkhoff averages, and verify $\lambda_{\text{BKL}} = \pi^2/(6\log 2) \approx 2.3731$ to four significant figures. (This is a textbook exercise; serves as sanity check.)

(F2) *Discretised Bianchi IX Krylov rate.* Take a truncation of the WDW (Wheeler–DeWitt) Hamiltonian on Bianchi IX in Misner variables to $n_{\text{rank}} = 10^3$ basis modes, compute the Lanczos coefficients b_n of a generic local field operator under the truncated modular evolution, and extract the Krylov-complexity growth rate λ_K^{trunc}.

(F3) *Rate comparison.* Conjecture 6 predicts $\lambda_K^{\text{trunc}} \leq 2\pi\,\lambda_{\text{BKL}} \approx 14.907$, with saturation in the BKL-chaotic regime and violation only in the (measure-zero) periodic-Kasner orbits. A numerical Lyapunov rate exceeding 14.907 by more than the truncation error would falsify Conjecture M45.1.

(F4) *Lemma A.1 compatibility.* Verify that the periodic-orbit subset of $X(2)$ (of σ-measure zero) maps to the eigenvalue-crossing set of paper #5 Lemma A.1 under the Misner-to-modular-surface change of variables.

8 Outlook

- *Lean formalization.* Step A (§4.1) and the Gauss-shift entropy computation (3) are tractable in mathlib (continued fractions and ergodic theory both exist). Steps B, C, D require von Neumann algebras (sparse in mathlib as of 2026).
- *Extension to Bianchi class A (types VIII, $VI_{-1/9}$).* The Manin–Marcolli framework extends [9] to all class A Bianchi types; the construction here should also extend.
- *Genus-$g \geq 2$.* The companion paper #4 (Modular Shadow LMP v2.5) noted that the genus-$g \geq 2$ extension fails for the FLRW case. The Bianchi IX setting is intrinsically tied to $X(2)$ (genus zero) and so the question is moot here; but extension to higher level $X(N)$ is interesting.
- *Co-authors targeted.* Matilde Marcolli (Caltech, for the symbolic-dynamics half) and Antony Speranza (Maryland, for the intrinsic-observer construction); see the companion `outreach_strategy.md`.

Discipline statement

- All 14 arXiv references in the bibliography below were WebFetch-verified on 2026-05-06 against `arxiv.org` directly.
- Six [**TBD: prove**] markers are flagged in §4–5; this paper proves *none* of the four conjectures M45.1.A–M45.1.D.
- No large-language-model output was treated as authoritative without independent live re-verification (Mistral strict-banned per project policy).
- This is a *conjecture paper*: the Manin–Marcolli correspondence (1) is a theorem, the Speranza/Witten/CLPW crossed-product framework is established, but the identification $\sigma_t^{\phi_\gamma} \leftrightarrow g_t$ (Conjecture 4) is the load-bearing open step.

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